

February 20, 2026

To: Steven Posnack, Principal Deputy Assistant Secretary for Technology Policy, ASTP/ONC
Department of Health and Human Services Mary E. Switzer Building, Mail Stop: 7033A 330
C Street SW, Washington, DC 20201

Re: Request for Information: HHS Health Sector AI RFI (Docket HHS-ONC-2026-0001)

Dear Mr. Posnack and the ASTP/ONC Review Team,

Thank you for the opportunity to provide feedback on the Department's approach to accelerating the safe adoption of AI in clinical care. I am commenting on the use of AI as an active agent in physical care—such as in autonomous surgical robotics, closed-loop infusion pumps, and automated bio-synthesis—I want to offer a perspective on the physical safety limits of software-only guardrails.

My primary feedback regarding the deployment of AI in physical clinical care is that software safety models, while necessary, are inherently probabilistic and subject to execution latency, kernel panics, and model deception. In high-consequence clinical environments, I recommend that HHS establish safety expectations rooted in physical hardware, rather than relying solely on software alignment.

Regarding Question 3, which touches on the novel legal and implementation issues relating to liability and accountability, the introduction of agentic AI creates a unique forensic challenge. Traditional medical liability relies on being able to audit what went wrong. However, if an AI-driven medical device acts unpredictably, the software logs it generates can be unreliable or spoofed if the system's root environment is compromised.

To resolve this accountability gap, HHS should encourage the adoption of hardware-resident event logging. If a device experiences a safety violation that results in a shutdown or a severed power connection, that event should generate a cryptographically signed record at the hardware level—completely independent of the AI's operating system. This acts as a true, unalterable "black box" for malpractice adjusters, hospital compliance boards, and the FDA, proving the physical state of the device regardless of what the software logs claim.

In response to Question 4 concerning AI evaluation methods and metrics, I suggest the Department rethink how it measures safety responsiveness. Currently, many robustness metrics focus on software response times. In physical care, such as robotic surgery, a software loop taking 20 to 50 milliseconds to recognize an error and issue a stop command can result in unintended tissue damage. HHS evaluation methods for physical AI tools should transition toward measuring "deterministic arrest windows." For instance, systems

should be evaluated on their ability to physically guarantee a zero-energy state at the actuator in under one or two milliseconds. Achieving this practically requires physical hardware interlocks (like a series interrupter on the power line) rather than just software-based stops.

Finally, regarding Question 5 and how HHS can best support private sector accreditation and certification, I believe the greatest impact would come from establishing baseline hardware states for clinical AI. One of the most effective ways to mitigate risk is to require that devices capable of causing physical harm default to a de-energized, normally-dark state. For example, power to drug-infusion pumps or surgical drives should be physically isolated until an independent hardware gate verifies the integrity of the clinical instruction set.

This concept extends to a variety of clinical applications. In automated molecular synthesis labs, reagent pumps should remain physically disconnected from power until the instruction sequence is verified to be free of prohibited pathogen sequences. Similarly, wearable or implantable AI controllers should have hardware disconnects that trigger if the device falls out of sync with the patient's natural biological rhythms.

Software recommendations will constantly evolve as AI models change, but the laws of physics do not. By standardizing the expectation that safety-critical interventions must happen at the physical energy-delivery layer, HHS can ensure that human clinicians—and not probabilistic software models—always retain the ultimate authority over patient care.

Thank you for your time and for the work your office is doing to secure the future of healthcare technology.

Sincerely,

Michael G. Bridges